

Intelligent Collaborative Workspace System

Optimizing IT Project Management through NLP and Innovative Features



Ziyu WANG	3035777547
Haowen ZHANG	3036580367
Ng Tsz Wing	3036372978
Cheng Yuen Chit	3036646915

Agenda

1. **Project Background**
2. **Project Objectives**
3. **Literature Review**
4. **Research Methodology**
5. **Findings**
6. **Recommendations**
7. **Conclusion**



1

Project Background

Project Background

In the rapidly evolving IT industry, effective project management is pivotal. Traditional manual assignment methods often lead to **inefficiencies** and **skill mismatches**.

This project leverages simulated job data to build an all-in one system that **simulates workflows with progress tracking, extracts skills using NLP**, and intelligently matches them to employee profiles.





2

Project Objectives

Project Objectives



Design & Collect

Design databases and identify/simulate data for core entities using object-oriented principles.



Develop Functions

Create a program library of functions for system operations and skill extraction.



Predictive Model

Implement a data mining model (Deep Learning) to relate project requirements to required skills



Visualize

Generate data visualizations to analyze project skill requirements and staff capabilities.

3

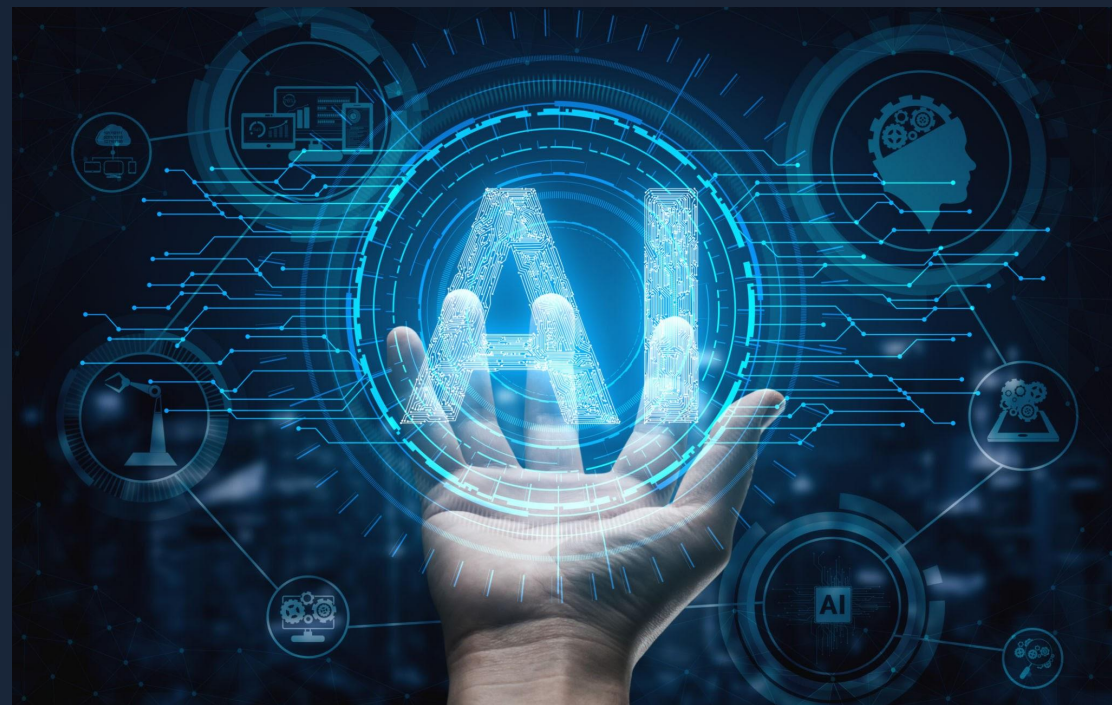
Literature Review

Literature Review: Skill Extraction

Extracting skills from unstructured text is foundational. Recent studies highlight hybrid approaches.

*"Najafi et al. (2023) propose a hybrid approach using **TF-IDF** and **BERT embeddings**, achieving 85% precision in extracting hard skills like Python and AWS ." for skills identification in IT job ads.*

This aligns with our use of **NLP for parsing task descriptions into actionable skill sets.**

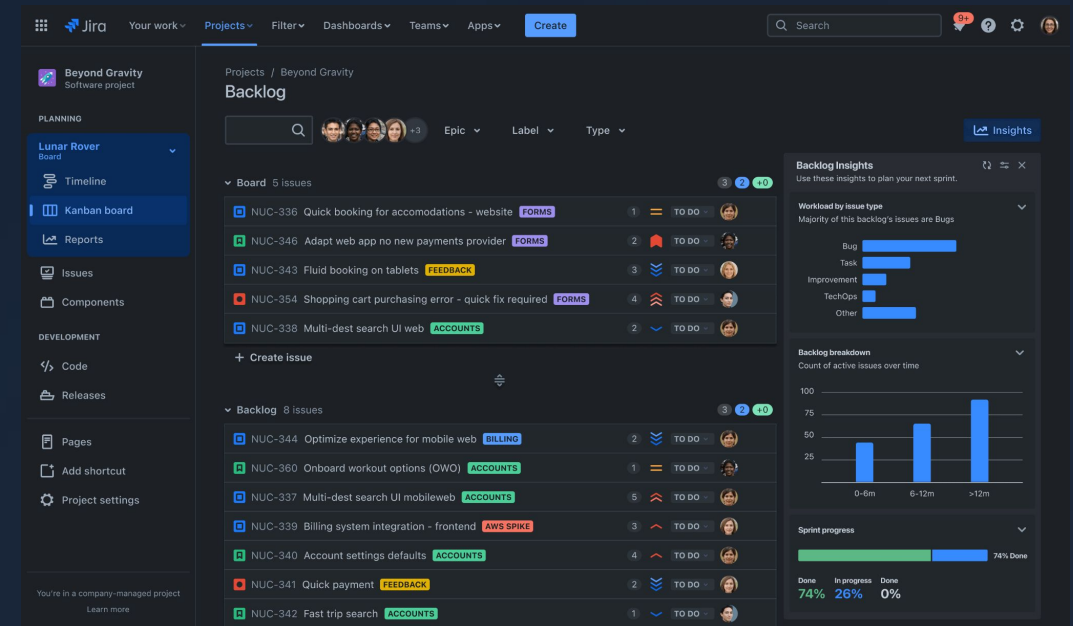


Literature Review: AI in Project Management System

Recent Studies demonstrates the importance of AI Application in project management system

*A system by Lee and Kim (2024) combines NLP with reinforcement learning for dynamic task assignment in agile teams, simulating JIRA-like workflows and yielding **20% faster project completions***

We also aware of the gaps of **limited integration of document versioning and notifications** in project management system which foster the development of our own prototype



4

Research Method

Research Methodology Overview



Data Sources Simulation

Simulate Employee
and Project Data (300
employees, 1000
projects) using Faker
library.



Modeling

Transformer-based
Skill Predictor
(DistilBERT) using
PyTorch



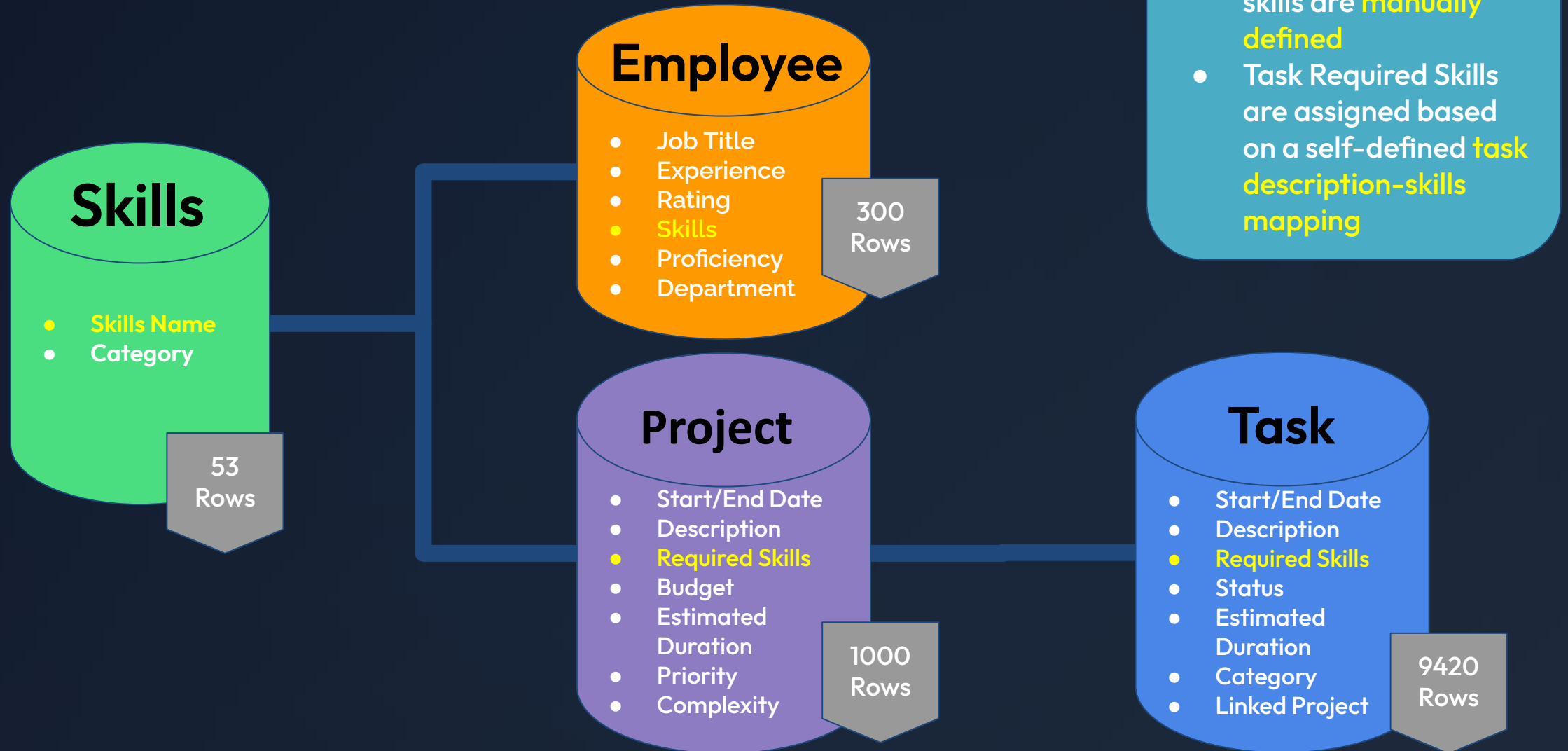
Comparison to existing system

Compare with other
existing collaborative
work system to see what
are the advantages of
our systems

5

Findings

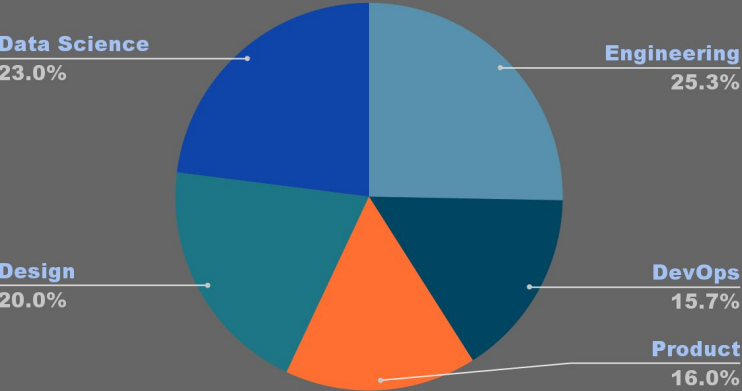
Data Simulation



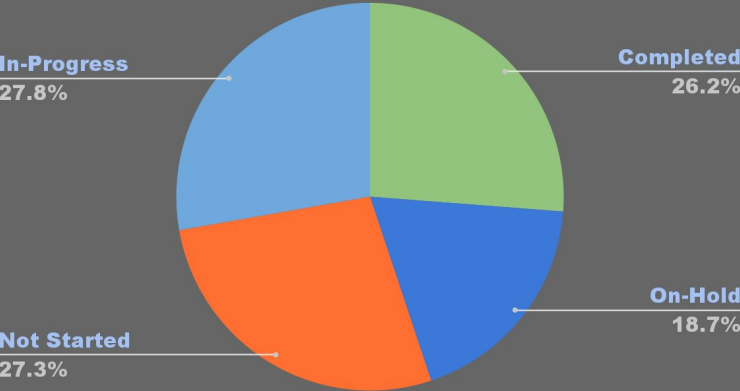
Findings

Simulated Data Analysis

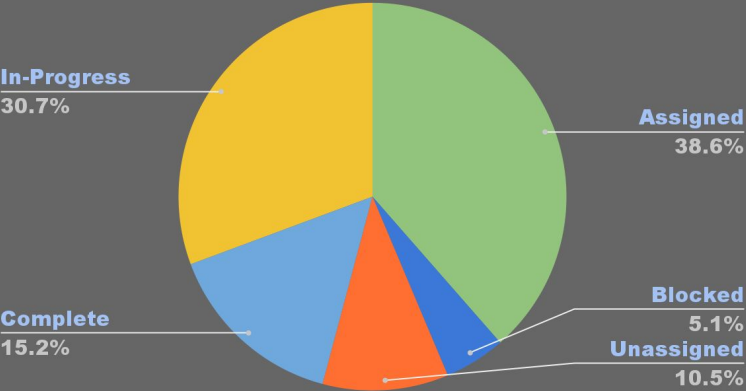
Employee Department Distribution



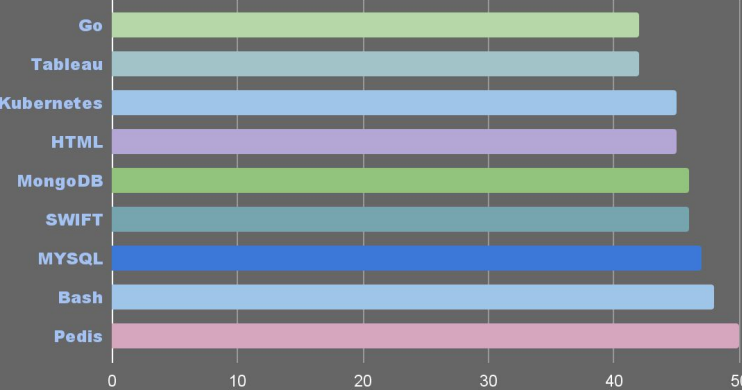
Project Status Distribution



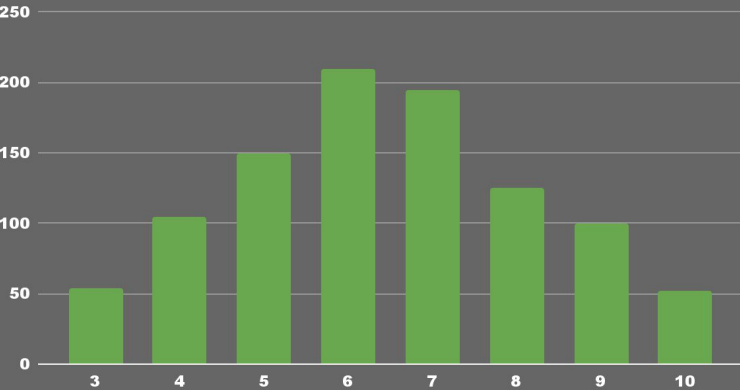
Task Status Distribution



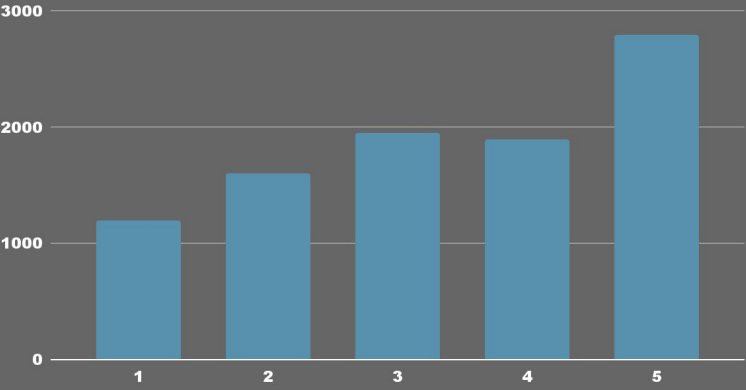
Top 10 Employee Skills



Project Complexity Distribution

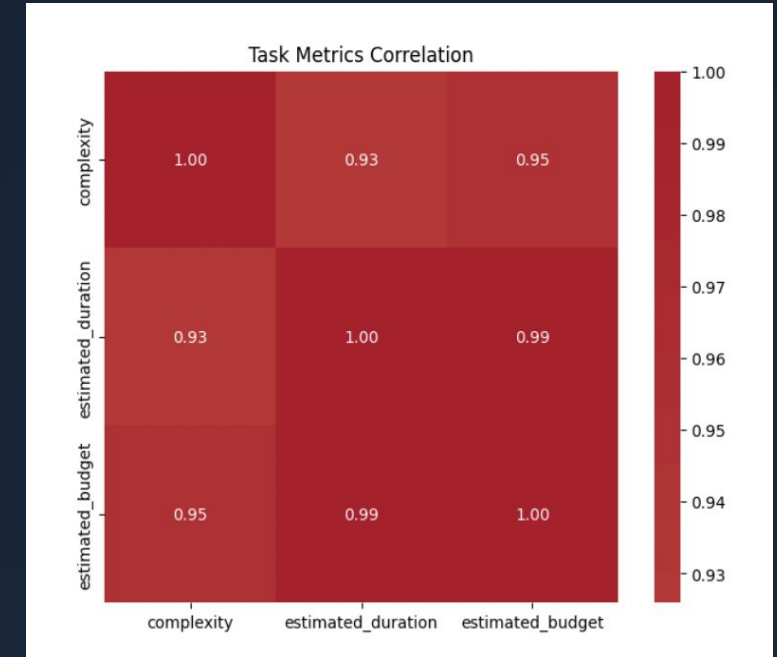
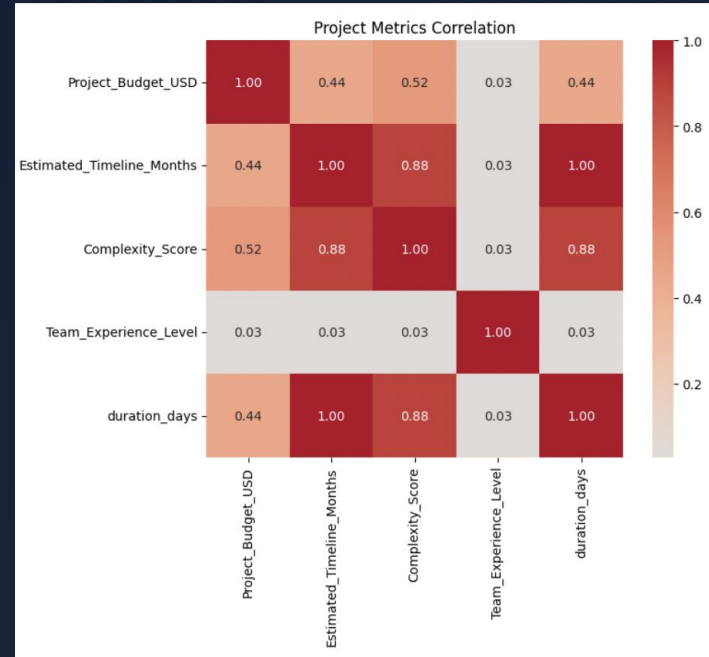
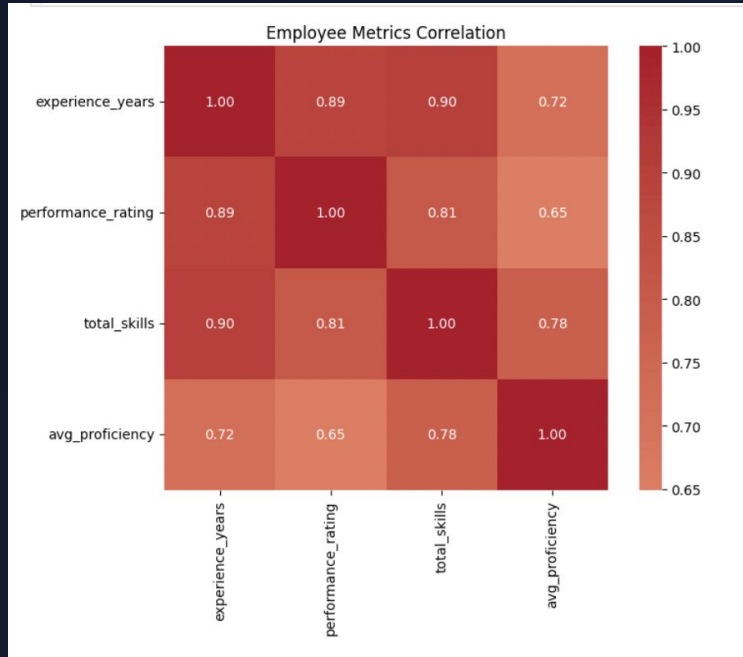


Task Complexity Distribution



Findings

Simulated Data Analysis



- **Employee's Performance rating** are highly correlate with **experience, number of skills and skills proficiency**
- Both **Budget/Time** of Project/Task are highly dependent of **Complexity** of Project/Task

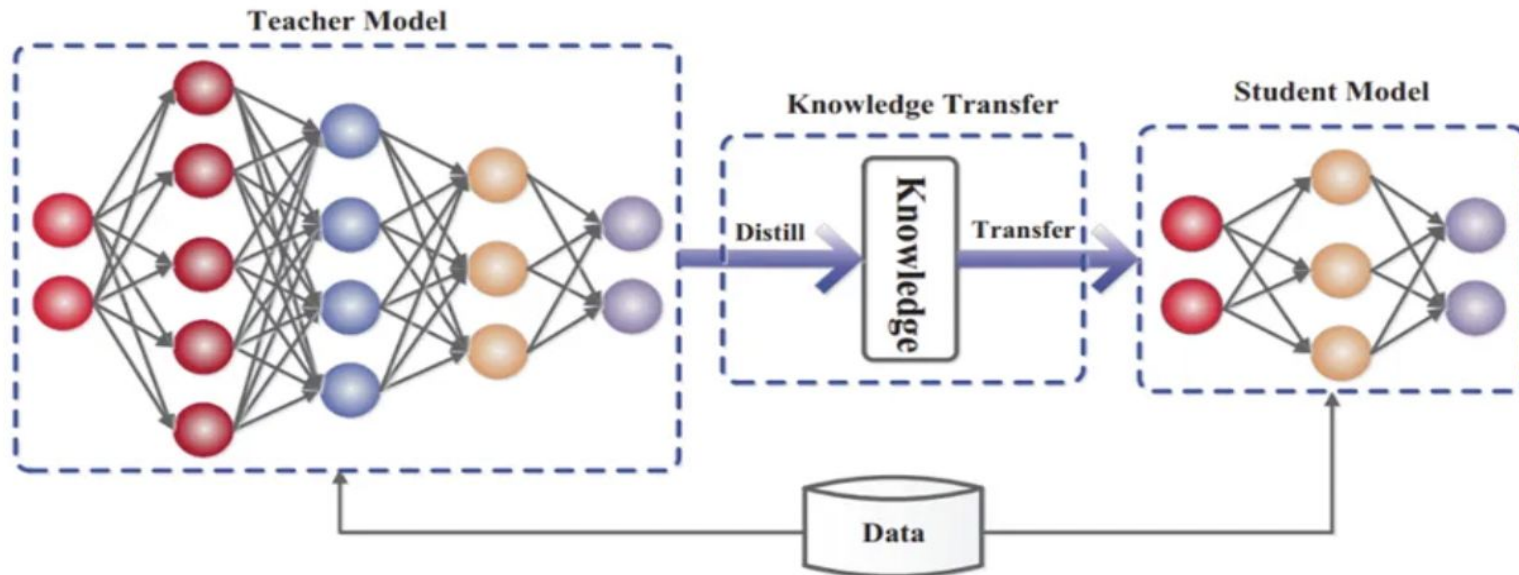
Findings

Modeling - DistilBERT

- DistilBERT stands for “Distilled BERT”
- A smaller, faster and lighter version of the BERT model

DistilBERT is trained using a compression technique called Knowledge Distillation:

- **Teacher (BERT):** The large, high-performing bidirectional transformer
- **Student (DistilBERT):** The compact model is trained to mimic the Teacher's output probabilities



Findings

Modeling - Preprocessing

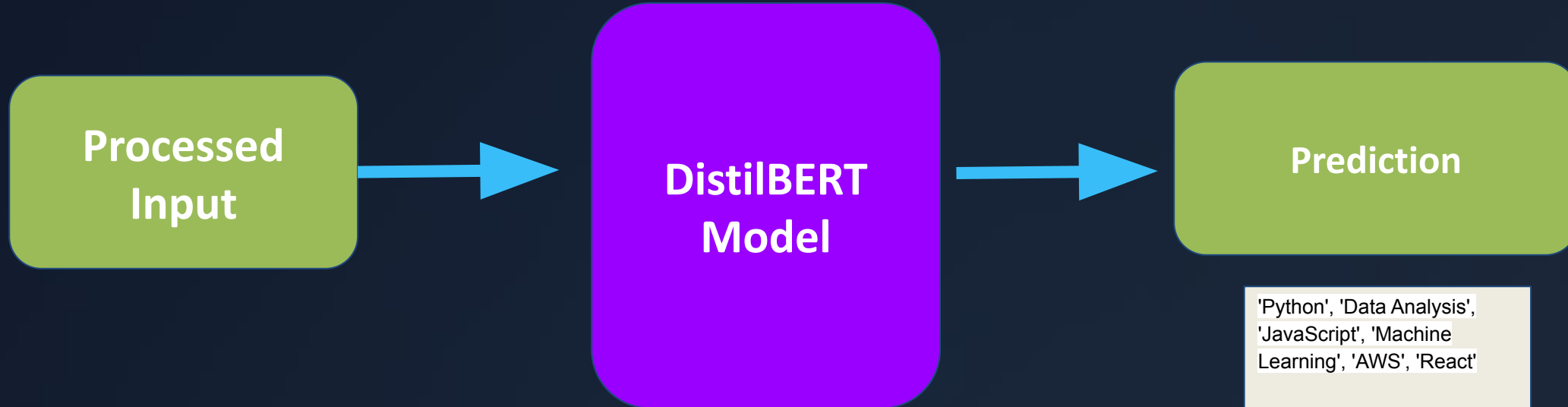


task_description	required_skillsets
Integrate enterprise-grade API gateway with re...	PHP; TensorFlow; MySQL
Create enterprise-grade file upload service us...	Leadership; PostgreSQL; Docker
Secure robust performance monitoring functiona...	Go; C#; PostgreSQL

1	1	0	0
0	0	1	0
1	0	0	0

Findings

Modeling - Training and Validation



Hyperparameters

- Epoches: 100
- Batch Size: 8
- Learning Rate: $3e-5$
- Optimizer: AdamW
- Loss: BCEWithLogitLoss

Findings

Modeling - Evaluation Metrics

1. Hamming Loss

$$HL = \frac{1}{N} \sum_{i=1}^N \frac{1}{L} \sum_{j=1}^L \mathbb{I}(y_{ij} \neq \hat{y}_{ij})$$

Where N is the number of Samples, L is the number of labels, y is the true/predicted labels. The lower the better

2. F1 Score

Final Validation Result

- Hamming Loss: 0.0
- F1 Score: 1.0

Findings

Feature Overview

Smart Dashboard

Centralized hub for progress tracking, status distribution metrics, and real-time system alerts.



Task AI

Automated workflow predictions for duration and budget based on task complexity scores.



Project Control

End-to-end setup wizards and integrated team allocation.



Collaborative Workspace



Employee Profiles

Detailed Employee profiles with skill matrices, proficiency tracking, and performance rating.



Data Analytics

Interactive Plotly visualizations for team capacity planning, performance review and project timeline with filter



Docs & Issues

Version-controlled file management and collaborative issue tracking threads.

Demo

Findings

Comparison with Existing Systems

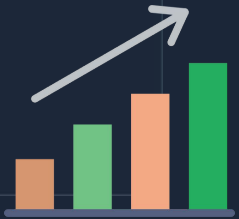
Advanced, Cost-Effective Skill Extraction

- Unlike the predefined tags in Monday.com or ClickUp's profile-based matching, our system uses **NLP** techniques
- Reduces setup time and improves accuracy



Integrated Modeling Tailored to IT

- We incorporate **domain specific predictions for task duration and budget using IT skills data**, going beyond Monday.com's general timelines or Jira's bottleneck alerts



Open-Source Flexibility and Zero Cost

- our system is **fully accessible and modifiable**—ideal for educational purposes or bootstrapped teams
- contrasting **the subscription models of competitors** (e.g., ClickUp's AI locked behind \$7+/user/month).



Focused IT Simulation Without Bloat

- It **simulates realistic IT scenarios** (e.g., agile breakdowns, skill gaps in cloud/devops) without the generalist overhead of tools like Jira, which require add-ons for full skill matching.



6

Recommendations

Recommendations

Innovative Feature - AI Team Recommendation

Feature

- AI recommend team members based on the project/task description

AI Recommendations

Top AI Recommendations based on skill matching:

Alice Johnson	Match: 83%	☒ Remove
Senior Software Engineer Engineering	Skills: Python, React, AWS	

Business Value

- **Reduction in Setup Time:** Automating skill matching reduces time in project initiation phase
- **Precision Matching:** Mitigates human error in overlooking niche skills required for complex IT tasks.

Recommendations

Innovative Feature - Predictive Budgeting

3. Predictions & Assignment

AI Predictions

Predicted Duration

10 days

Predicted Budget

\$2,000

Estimated Duration (Days)*

10

-

+

Estimated Budget (USD)*

2000

-

+

Recommendation based on 'Frontend' (Complexity: 3):

- Estimated Duration: 10 days
- Estimated Budget: \$2,000

Feature

- Predict the task duration and budget based on task complexity by LinearRegression Model

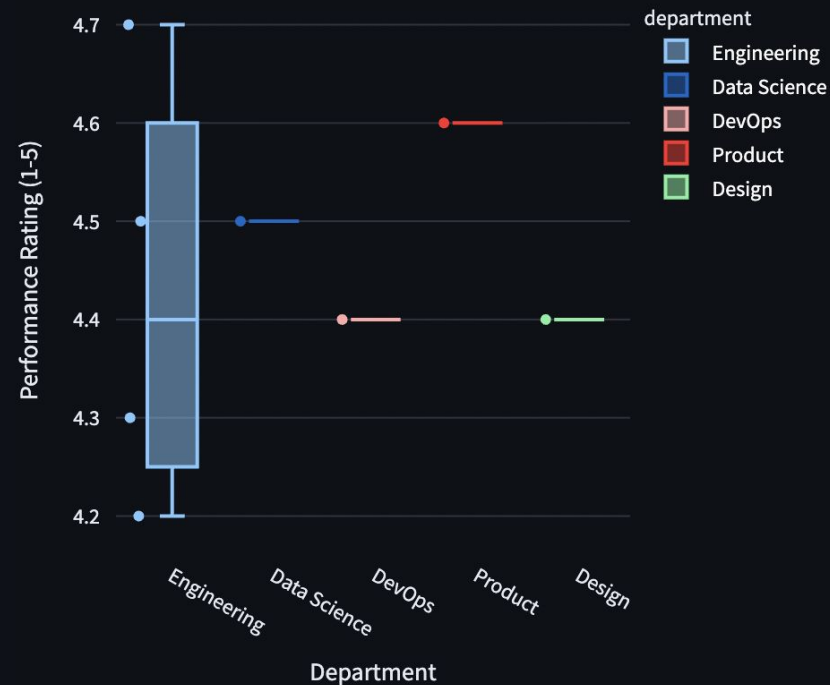
Business Value

- **Financial Accuracy:** Linear regression models provide data-backed baselines, reducing budget overruns.
- **Resource Planning:** Complexity-based duration estimates allow for tighter, more reliable roadmap scheduling.

Recommendations

Innovative Feature - Performance Benchmarking

Performance Rating by Department



Feature

- Rating Distribution of Departments

Business Value

- **Objective Performance Overview:** Provides senior management with a macro-view of departmental health. Helps isolate high-performing units for reward and identifying departments needing training interventions.

Recommendations

Innovative Feature - Workload Balancing



Feature

- Real-time visualization of task distribution per member

Business Value

- **Reduce turnover cost:** Prevents employee burnout and identifies underutilized resources. This retention strategy reduces high turnover costs associated with overworked IT staff.

7

Conclusion

Conclusion

We successfully designed and prototyped an intelligent system for IT task assignment.

- **Automation:** All-In-One Collaboration tools for faster and better project management
- **Accuracy:** The Deep Learning model demonstrated high fidelity in matching requirements to skills.
- **Impact:** The solution reduces the non-billable hours and overhead costs associated with project mismanagement.



"The system shifts the organization from intuitive management to evidence-based decision making."

Recommendations

Business Insight



AI Team Recommendation

Feature: Real-time visualization of task distribution per member.

- ✓ Reduction in Setup Time: Automating skill matching reduces time in project initiation phase

Scope Expansion

Generalizing the model to other industries beyond IT and implementing multi-user authentication for enterprise-scale security.

Research Methodology Overview



Data Sources Simulation

Simulate Employee and Project Data (300 employees, 1000 projects) using Faker library.



Modeling

Transformer-based Skill Predictor (DistilBERT) using PyTorch



Innovative Features

Develop with innovative Features such as Department Performance Analysis and employee's workload distribution



Comparison to existing system

Compare with other existing collaborative work system to see what are the advantages of our systems

Matching Algorithms & Gaps

Matching Algorithms

Cosine similarity on skill vectors is widely advocated (Balog et al., 2019). Advanced methods use Graph Neural Networks (GNNs) for team dynamics (Gupta & Bhattacharya, 2022).

Identified Gaps

While existing works excel in isolated matching, few integrate full project lifecycles with visualizations. This project bridges that gap with an end-to-end prototype.

Project Background

Problem Statement

- **Inefficiency:** Manual assignment processes are slow and error-prone.
- **Skill Mismatches:** Static inventories fail to capture dynamic IT needs (e.g., Cloud, AI).
- **Resource Waste:** Mismatched skills lead to project delays and overburdened staff.
- **Integration Gap:** Lack of tools combining employee profiles, and project structures.



8

Future Work

Future Work

Technical Enhancements

Integration with real-time enterprise tools (Jira, Slack) and deployment of advanced features like deadline notifications.

Scope Expansion

Generalizing the model to other industries beyond IT and implementing multi-user authentication for enterprise-scale security.

Q & A

Appendix: Evaluation Metric

The F1-Score used for evaluation is the harmonic mean of Precision and Recall, crucial for handling imbalanced multi-label data.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$